

Materials Optimization of an EIT-Like Terahertz Metasurface: Metal Thickness, Substrate Selection, and Coupled-Resonator Analysis

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Abstract—We report a full-wave HFSS study of the EIT-like terahertz metasurface of Zhang et al. [1], targeting two fabrication parameters not systematically varied in the original work: aluminum film thickness relative to the THz skin depth, and substrate dielectric properties. Simulations will sweep six metal thicknesses and four thin-film substrates; a coupled-resonator model will be fitted at each simulation point to extract γ_1 , γ_2 , κ , and δ , connecting full-wave results directly to the device physics. This report presents the completed Phase 1 baseline validation.

Index Terms—EIT metasurface, terahertz, coupled resonators, skin depth, group delay, HFSS, substrate optimization.

I. INTRODUCTION

Electromagnetically induced transparency (EIT) in THz metasurfaces is produced by destructive interference between two bright resonator modes, yielding a narrow transmission window with the steep phase dispersion needed for slow-light and sensing applications [2], [3]. Zhang et al. [1] realized a graphene-integrated aluminum-on-polyimide version (GrE & MS) with a transparency window at 0.87 THz, dips at 0.70 and 1.19 THz, a peak group delay of 5.5 ps, and a 182% graphene-controlled modulation depth near the Dirac point [4], [5].

The paper, however, fixes both the metal and the substrate without justification. At 1 THz the skin depth of aluminum is $\delta_s \approx 84$ nm (using $\delta_s = \sqrt{2/\mu_0\sigma\omega}$ with $\sigma = 3.56 \times 10^7$ S/m), yet the film is 200 nm— $\sim 2.4\times$ deeper than needed for opacity. Singh et al. [6] showed resonance quality in THz metamaterials saturates near δ_s , so thinner films may preserve performance while reducing deposition time and stress. On the substrate side, the polyimide (PI) loss tangent ($\tan \delta = 0.05$) is among the highest of available thin-film polymers; Parylene-C and BCB offer loss tangents $10\text{--}17\times$ lower at comparable permittivity [8], which should narrow the EIT linewidth and increase group delay. Thick substrates ($\geq 300\text{ }\mu\text{m}$ Si) are not considered because Fabry–Pérot modes corrupt the EIT window at sub-mm wavelengths [7]. The graphene modulation layer is intentionally excluded from the parametric study: graphene controls dynamic tuning via Fermi-level shifting but does not determine the structural resonance frequencies, linewidths, or group delay that define the passive EIT performance being optimized here. This work fills both gaps with a controlled HFSS parametric study of the passive

metasurface tied to a coupled-resonator analytical model, yielding fabrication guidelines grounded in device physics.

II. METHODOLOGY

A. Phase 1 — Baseline Validation

The HFSS unit cell replicates the passive metasurface of [1], omitting the graphene layer to isolate the structural parameters under study: period $P_x = P_y = 100\text{ }\mu\text{m}$; inner-arc radius $R_1 = 34\text{ }\mu\text{m}$; inner-arc radii $R_2 = R_3 = 10\text{ }\mu\text{m}$, $R_4 = 15\text{ }\mu\text{m}$; conductor width $w = 5\text{ }\mu\text{m}$; aluminum ($\sigma = 3.56 \times 10^7$ S/m) at 200 nm on $5\text{ }\mu\text{m}$ PI ($\epsilon_r = 3.1$, $\tan \delta = 0.05$). Floquet-port excitation with primary/secondary periodic boundaries yields S_{21} from 0.5 to 1.5 THz; the maximum Delta S convergence criterion is set to $\Delta S < 0.02$. The coupled-resonator susceptibility [1], [7]

$$\chi \propto \frac{(\omega - \omega_0 - \delta) + i\frac{\gamma_2}{2}}{(\omega - \omega_0 + i\frac{\gamma_1}{2})(\omega - \omega_0 - \delta + i\frac{\gamma_2}{2}) - \frac{\kappa^2}{4}} \quad (1)$$

is fitted to $|S_{21}|^2$ via a differential-evolution global optimizer to extract γ_1 , γ_2 , κ , δ .

B. Phase 2 — Metal Thickness Sweep

Aluminum thickness is varied at 50, 100, 150, 200, 300, and 500 nm with all other parameters fixed. Both the EIT quality factor and the group delay

$$\tau_g = -\frac{d\phi(\omega)}{d\omega} \quad (2)$$

are expected to plateau once the film exceeds δ_s [6]. Re-fitting Eq. (1) at each thickness tracks how $\gamma_{1,2}$ evolve with film opacity and identifies the minimum thickness that preserves EIT performance.

C. Phase 3 — Substrate Sweep

At the Phase 2 optimum thickness, the substrate is replaced across the four candidates in Table I at constant $5\text{ }\mu\text{m}$. Permittivity approximately scales the resonance frequency as $f_0 \propto \epsilon_{\text{eff}}^{-1/2}$ [3]; loss tangent increases the non-radiative damping contribution to $\gamma_{1,2}$ through dielectric dissipation. The optimum maximizes $Q \times T_{\text{peak}} \times \tau_g$; note that Q and τ_g are correlated in EIT systems, so this metric primarily weights slow-light and transmission quality jointly.

TABLE I: Thin-film substrate candidates. THz parameters from [8].

Material	ε_r	$\tan \delta$	Note
Polyimide (PI)	3.10	0.050	Baseline [1]
Parylene-C	2.95	0.003	$17\times$ lower loss
BCB	2.67	0.005	Lower ε_r & loss
SU-8	3.90	0.040	Higher ε_r

III. PHASE 1 RESULTS

A. Transmission

Figure 1 and Table II compare the HFSS baseline to [1]. Dip 1 reproduces to within 0.7%, confirming bright mode 1 is correctly captured. The window and dip 2 are red-shifted by 7–11%, consistent with the bulk PI permittivity ($\varepsilon_r = 3.1$) assigned in HFSS being slightly higher than the effective value of a $5\mu\text{m}$ spin-coated film [2], [3]. The internal frequency ratios and lineshape are self-consistent; since Phases 2 and 3 report results *relative* to this baseline, the absolute offset does not affect the comparative conclusions.

TABLE II: Spectral features: Zhang et al. [1] vs. HFSS baseline.

Feature	Ref. [1]	HFSS	Δ
Dip 1	0.70 THz	0.705 THz	+0.7%
Window	0.87 THz	0.81 THz	−6.9%
Dip 2	1.19 THz	1.06 THz	−10.9%
Peak τ_g	5.5 ps	5.3 ps	−3.6%

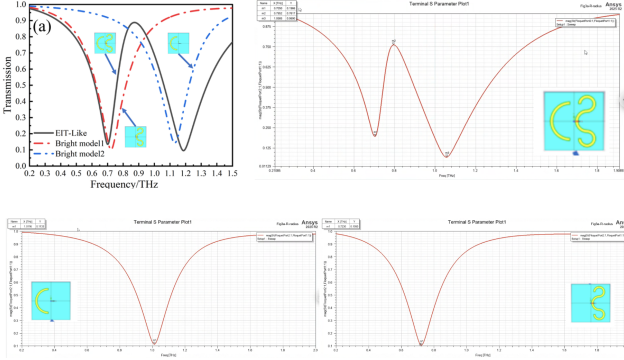


Fig. 1: Transmission spectra. *Left*: Fig. 3(a) of [1]—EIT-like (solid), bright-mode 1 (red dash-dot), bright-mode 2 (blue dash-dot). *Right/bottom*: HFSS $|S_{21}|$ for the full GrE & MS cell and isolated resonators; dips at 0.705 and 1.06 THz, window at 0.81 THz.

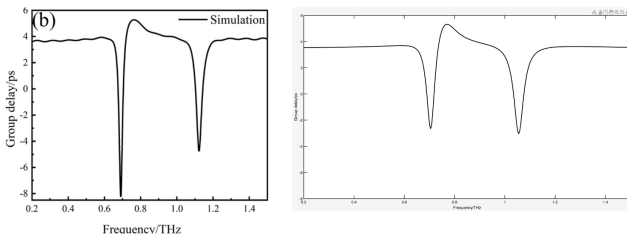


Fig. 2: Group delay τ_g . *Left*: Reference [1], peak ≈ 5.5 ps. *Right*: HFSS, peak ≈ 5.3 ps.

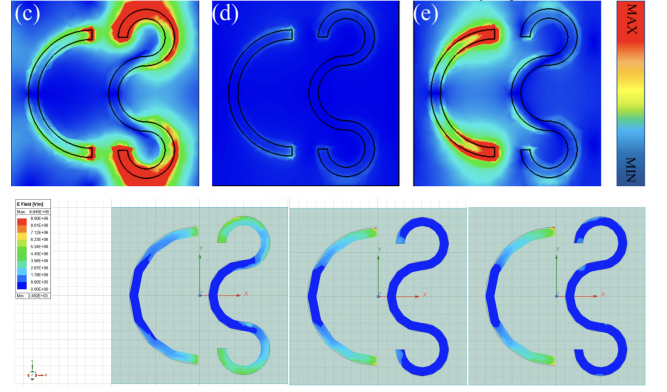


Fig. 3: Electric field distributions. *Top*: $|E|$ cross-sections from [1] at 0.70, 0.87, and 1.19 THz; gap-field hotspots are concentrated on the small S-arc (bright mode 1) at dip 1, fields are suppressed at the window, and hotspots transfer to the large C-arc (bright mode 2) at dip 2. *Bottom*: HFSS conductor-surface $|E|$ at 0.706, 0.81, and 1.10 THz on a shared scale.

B. Group Delay

The simulated τ_g (Fig. 2) peaks at 5.3 ps, 3.6% below the reference value of 5.5 ps, with the expected positive peak at the window and negative troughs at the resonance dips. The simulated troughs reach ≈ -4 ps versus ≈ -8 ps in the reference simulation.

C. Electric Field Maps

Figure 3 places the reference and HFSS field maps side by side. The reference cross-sections clearly show the mode-switching signature of EIT—hotspots on the small S-arc (bright mode 1) at dip 1 (0.70 THz), near-zero fields across both resonators at the window, and hotspot transfer to the large C-arc (bright mode 2) at dip 2 (1.19 THz). The HFSS maps, plotted on the conductor surface with a shared color scale, do not resolve this contrast thoroughly.

D. Coupled-Resonator Fit

Eq. (1) was fitted to the HFSS $|S_{21}|^2$ data over 0.50–1.30 THz using a differential-evolution global optimizer, with a slowly-varying background term added to account for the frequency-dependent substrate transmission. The resulting parameters are: $\gamma_1 = 0.126$ THz, $\gamma_2 = 0.296$ THz, $\kappa = 0.096$ THz, $\delta = 0.440$ THz. Figure 4 shows the overlay of the fit against the HFSS data. The fit reproduces both resonance dips and the transparency window; the window-peak amplitude is slightly underestimated because the analytical model assumes a flat background, whereas the metasurface has a frequency-dependent diffraction loss. The ordering $\gamma_1 < \gamma_2$ indicates that the small S-arc resonator (mode 1, dip 1 at 0.70 THz) has a narrower linewidth than the large C-arc resonator (mode 2, dip 2 at 1.06 THz); the larger C-arc has a longer current path and higher radiative loss, consistent with $\gamma_2 > \gamma_1$. In Phase 2, $\gamma_{1,2}$ are expected to decrease as film thickness approaches δ_s , saturating once the metal is opaque. In Phase 3, $\gamma_{1,2}$ will increase with substrate $\tan \delta$ while κ will shift with ε_r via dielectric screening, providing a direct

analytical handle on how substrate choice modifies window width and group delay.

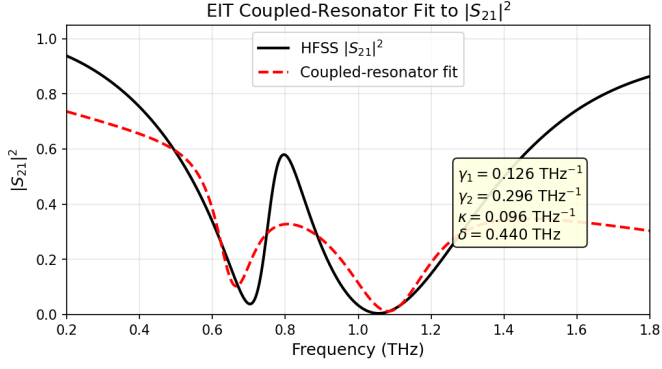


Fig. 4: Coupled-resonator fit (red dashed) overlaid on the HFSS $|S_{21}|^2$ baseline (black solid).

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